Course Two

Get Started with Python

Instructions

Use this PACE strategy document to record decisions and reflections as you work through this end-of-course project. You can use this document as a guide to consider your responses and reflections at different stages of the data analytical process. Additionally, the PACE strategy documents can be used as a resource when working on future projects.

Course Project Recap

Regardless of which track you have chosen to complete, your goals for this project are:

- [x] ~~Complete the questions in the Course 2 PACE strategy document~~
- [x] ~~Answer the questions in the Jupyter notebook project file~~
- [x] ~~Complete coding prep work on project's Jupyter notebook~~
- [x] ~~Summarize the column Dtypes~~
- [x] ~~Communicate important findings in the form of an executive summary~~

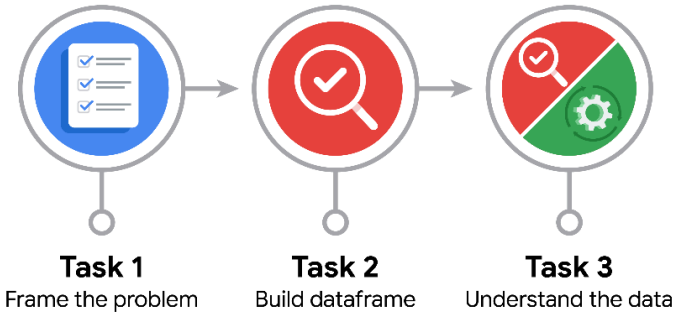
Relevant Interview Questions

Completing the end-of-course project will help you respond these types of questions that are often asked during the interview process:

- Describe the steps you would take to clean and transform an unstructured data set.
- What specific things might you look for as part of your cleaning process?
- What are some of the outliers, anomalies, or unusual things you might look for in the data cleaning process that might impact analyses or ability to create insights?

Reference Guide

This project has three tasks; the visual below identifies how the stages of PACE are incorporated across those tasks.



Data Project Questions & Considerations

PACE: Plan Stage

- How can you best prepare to understand and organize the provided information?

First, I will clarify the primary objective of the project (predicting **user churn**). Next, I will examine the **data dictionary** to analyze what each variable represents (e.g., sessions, driving_days, label). I will then adopt an exploratory perspective to understand the structure of the **missing values** (700 rows) and evaluate whether the dataset is sufficient to address the core research question of the project.

- What follow-along and self-review codebooks will help you perform this work?

The Python Notebook (Lab) provided within the scope of the project will serve as my primary guide. Specifically, the **pandas** documentation is critical for data manipulation, while **NumPy** is essential for statistical calculations. Furthermore, I will utilize previous similar **EDA (Exploratory Data Analysis)** templates and the project's **PACE strategy document** as self-review tools to validate the accuracy of my workflow.

- What are some additional activities a resourceful learner would perform before starting to code?

Before writing any code, I establish hypotheses, such as "Is there a significant difference between iPhone and Android users?" or "Do professional drivers churn at a higher rate?" Additionally, I consider how **outliers** might distort the analysis results and plan why I should utilize the **median** instead of the **mean**. Finally, I pre-determine which visualization types (e.g., bar charts, boxplots) will be most effective for the **Executive Summary** to be presented to stakeholders.

PACE: Analyze Stage

- Will the available information be sufficient to achieve the goal based on your intuition and the analysis of the variables?

The current dataset, consisting of **13 columns and 14,999 rows**, provides a solid foundation for understanding user behavior and predicting **churn**. Specifically, variables such as `driving_days`, `sessions`, and `total_km` appear to be directly correlated with churn. While the **700 missing values** in the `label` column are randomly distributed and may not significantly bias the analysis, incorporating qualitative data—such as user feedback regarding reasons for churn—or completing these missing entries could be beneficial for achieving more precise predictions in the future.

- How would you build summary dataframe statistics and assess the min and max range of the data?

I will generate a summary table containing the mean, standard deviation, minimum, maximum, and **quartile** values for all numerical variables using the `df.describe()` method in Python. By examining the **range** between the minimum and maximum values, I will assess the data's dispersion. For instance, if the maximum driving distances are significantly far from the mean, it indicates the presence of **outliers** in the data. This confirms why I should focus on the **median** rather than the mean to accurately understand the "typical user.

- Do the averages of any of the data variables look unusual? Can you describe the interval data?

The unusually high average daily driving distance (~700 km) among churned users is a significant finding; this suggests that this segment may consist of **professional drivers**, such as long-haul truckers. In terms of **interval data**, the number of days users actively drive with the app (`driving_days`) ranges from 0 to 30 days. The fact that churned users have fewer driving days but a higher driving intensity (duration per kilometer) indicates that the data within this interval is not **homogeneously distributed**.

PACE: Construct Stage

Note: The Construct stage does not apply to this workflow. The PACE framework can be adapted to fit the specific requirements of any project.

PACE: Execute Stage

- Given your current knowledge of the data, what would you initially recommend to your manager to investigate further prior to performing exploratory data analysis?

I would recommend that my manager investigate the exceptionally high average daily driving distances (~700 km), particularly within the **churned user group**. This data suggests that these users might be **commercial or professional drivers** (e.g., logistics, taxi services) rather than individual commuters. Before proceeding with the **EDA (Exploratory Data Analysis)** phase, it is critical to verify how well this "professional driver" segment aligns with our **business objectives** and to determine whether these **outliers** stem from data collection errors or represent genuine user behavior.

- What data initially presents as containing anomalies?

Initial inspections reveal significant anomalies in the total_km and duration_minutes variables. The fact that some users appear to have covered impossibly high distances within very short durations (or vice versa) confirms the presence of outliers in the dataset. Furthermore, the 700 missing values in the label column (churn status) should be treated as an anomaly regarding the integrity of the analysis; it is essential to investigate whether this missingness is concentrated within a specific user segment, such as Android users.

- What additional types of data could strengthen this dataset?

To strengthen the dataset, the following additional data would be highly beneficial:

- **User Feedback:** Survey results indicating the reasons why users uninstall the app.
- **Location Data:** Coarse location information to understand whether churn occurs in specific regions (e.g., urban vs. intercity).
- **App Performance Metrics:** Data on app crash rates or loading speeds, which would allow us to measure the impact of technical issues on churn.
- **Timestamps:** Knowing the specific times of day when users drive would help us solidify the distinction of the "professional driver" segment.